



Predicting voter turnout and election outcomes at the precinct level



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Abstract

In the United States, polling data are collected at the state level, but **election results vary substantially within states** (e.g., between rural and urban areas). Unlike polls, election results are often available at the precinct level, and these fine-scale data can provide insight into how primary elections, general elections, and voter turnout are related at a granular level. Using **primary and general data** at the precinct level for past presidential races, we aim to explore a voter dynamics model that quantifies turnout behaviour, contributing to a deeper understanding of electoral prediction beyond polling.

Background

Election forecasters often rely on polling data, economic indicators, and demographic information typically available only at the state or national level. However, primary election results provide political sentiment at the community level, as each precinct has only a few thousand eligible voters.

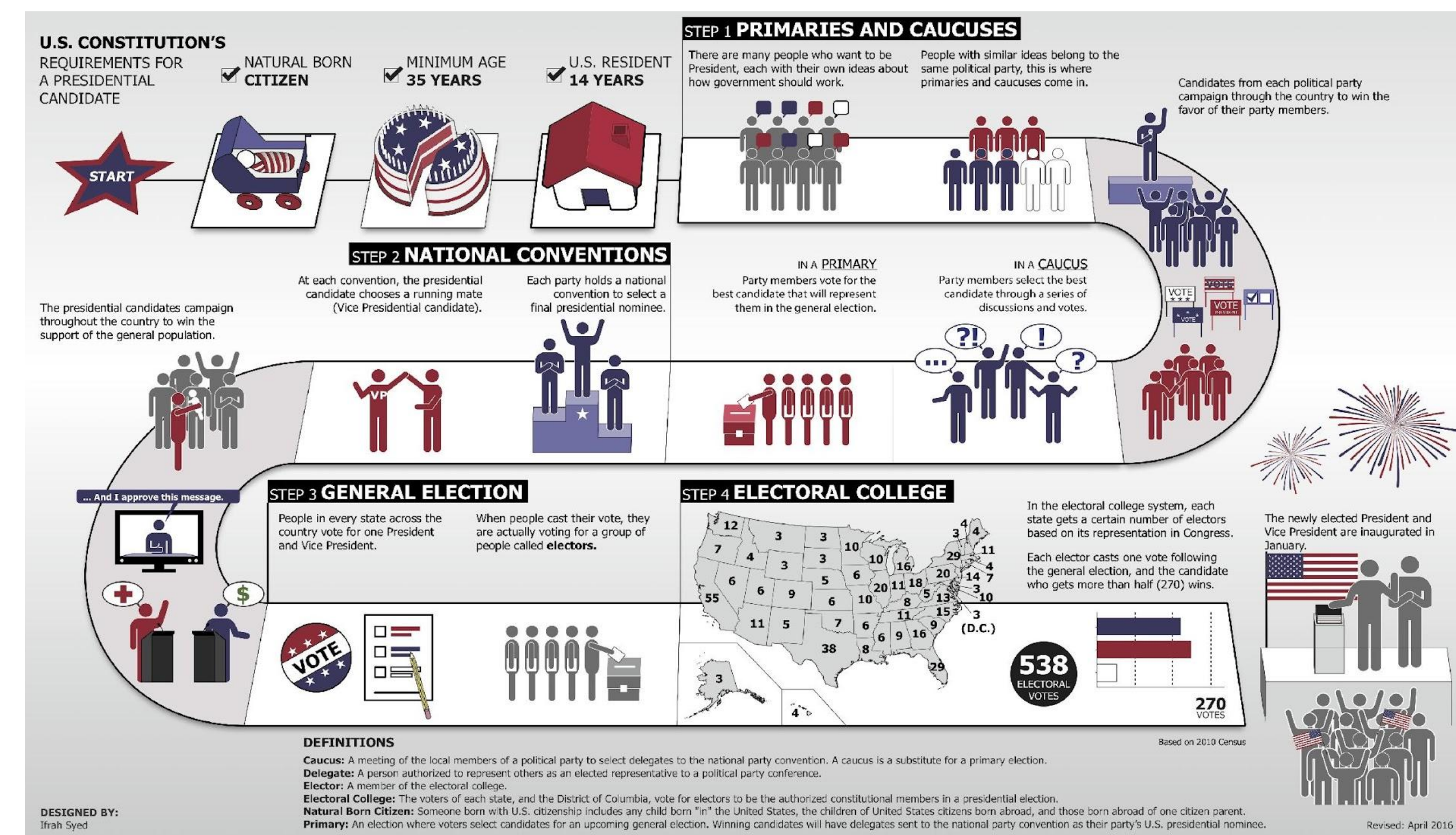
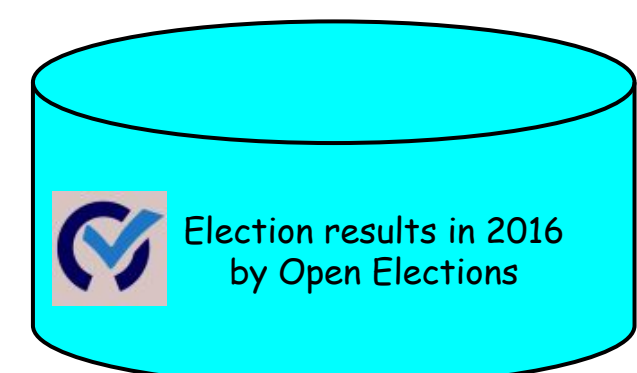


Figure 1: Overview of the US presidential election process. In brief, candidates for president must first be nominated by members of their political party (e.g., Republicans or Democrats) following primary elections in each state, which occur January through June in an election year. The nominated candidates then compete in the general election, which occurs in November.

Dataset



#ID	Precinct	pri_dem.Clinton	pri_dem.Sanders	pri_rep.Trump	pri_rep.Kasich	gen_dem.Clinton	gen_rep.Trump
1	PRECINCT.A	127	99	320	150	480	480
2	PRECINCT.B	243	112	270	80	210	350
3	PRECINCT.C	198	145	310	102	180	375
:	:	:	:	:	:	:	:

Table 1: Example structure of precinct-level dataset from the 2016 elections.

Research Questions

- What is the relationship between precinct-level presidential **primary results** and general election results?
- What is the relationship between **party consensus** (as measured by candidate ideology and party entropy) during the election cycle and the general election results?
- What publicly available features at the precinct level are the **best predictors** of presidential election **turnout and results**?



Github

Null Model

A **null model** for general election results based only on primary election results is:

$$\text{Vote share in general} = \text{Vote share in primary}$$

However, the winning party in the general election tends to underperform the null model, especially in high-turnout precincts.

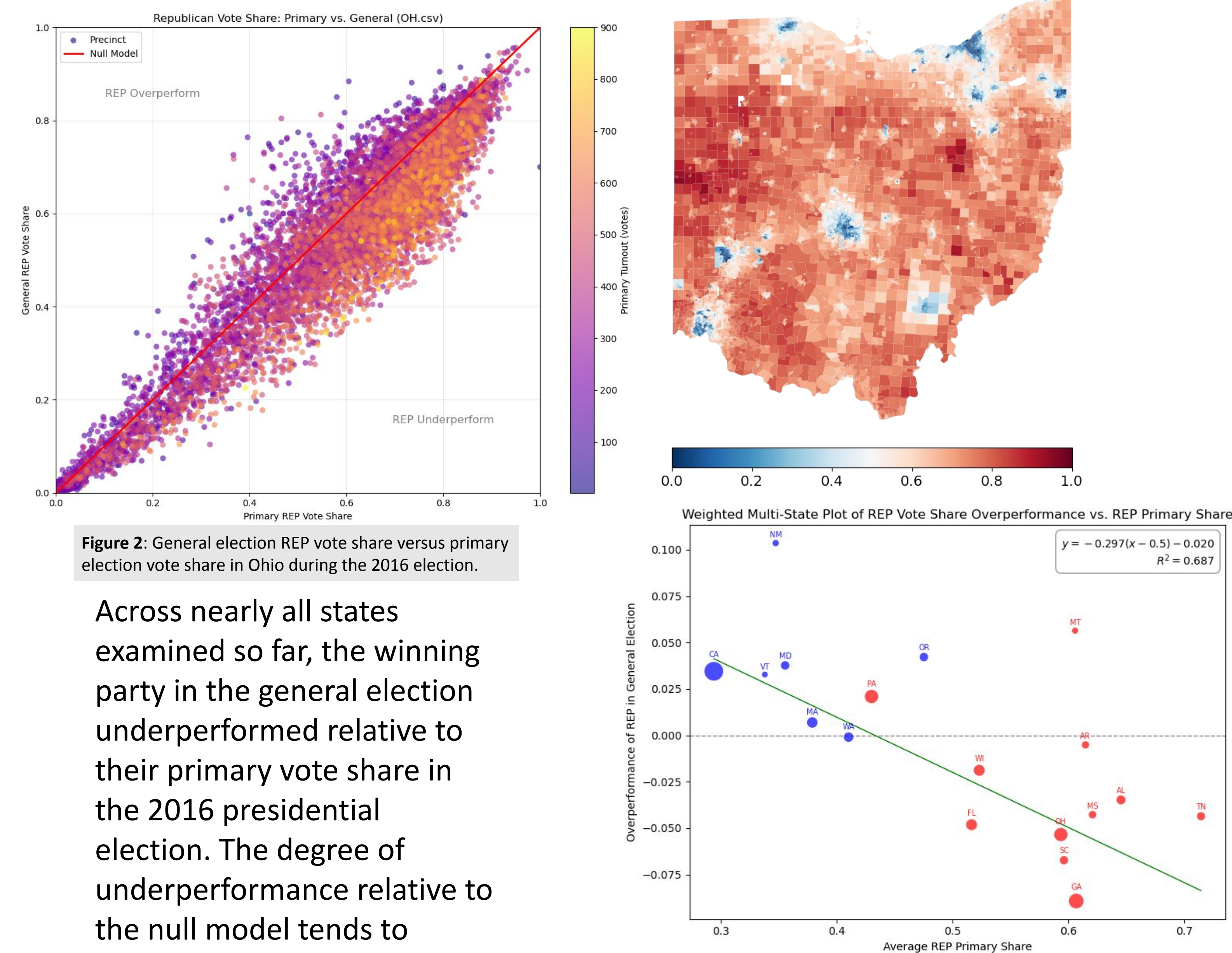


Figure 2: General election REP vote share versus primary election vote share in Ohio during the 2016 election.

Across nearly all states examined so far, the winning party in the general election underperformed relative to their primary vote share in the 2016 presidential election. The degree of underperformance relative to the null model tends to increase as the winning margin increases.

Figure 3: Weighted multi-state plot of Republican overperformance in the 2016 general election versus Republican primary share. Red states indicate a Republican general election win, and circle size reflects the number of general voters.

Complacency Effect Hypothesis

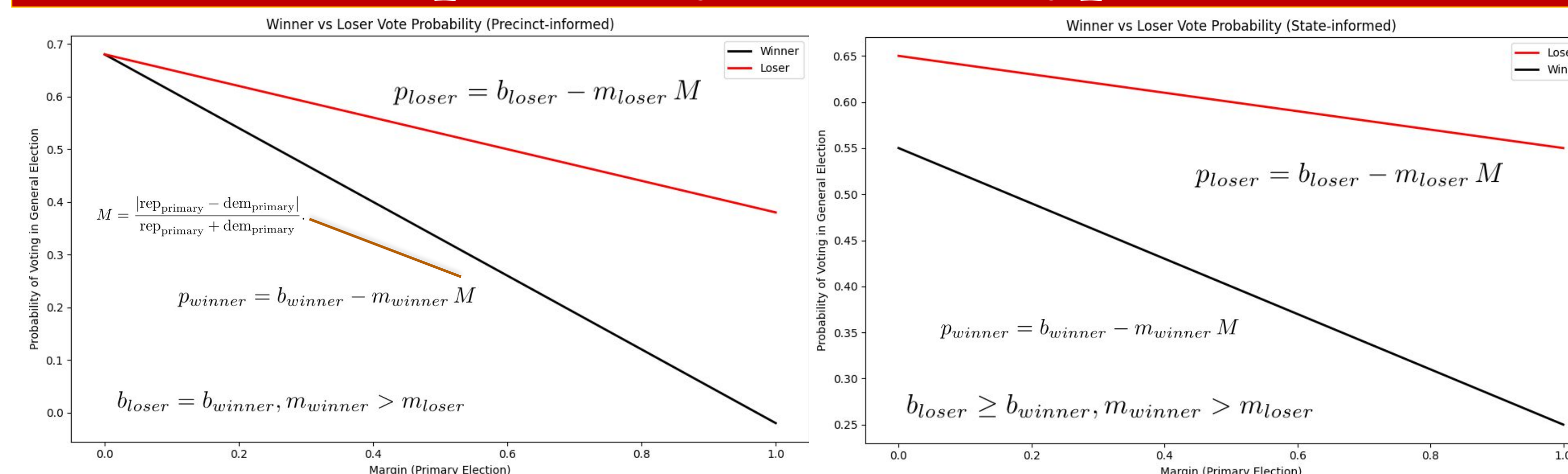


Fig 4a: Probability of voting for registered voters where both parties remain competitive by the result at the precinct level.

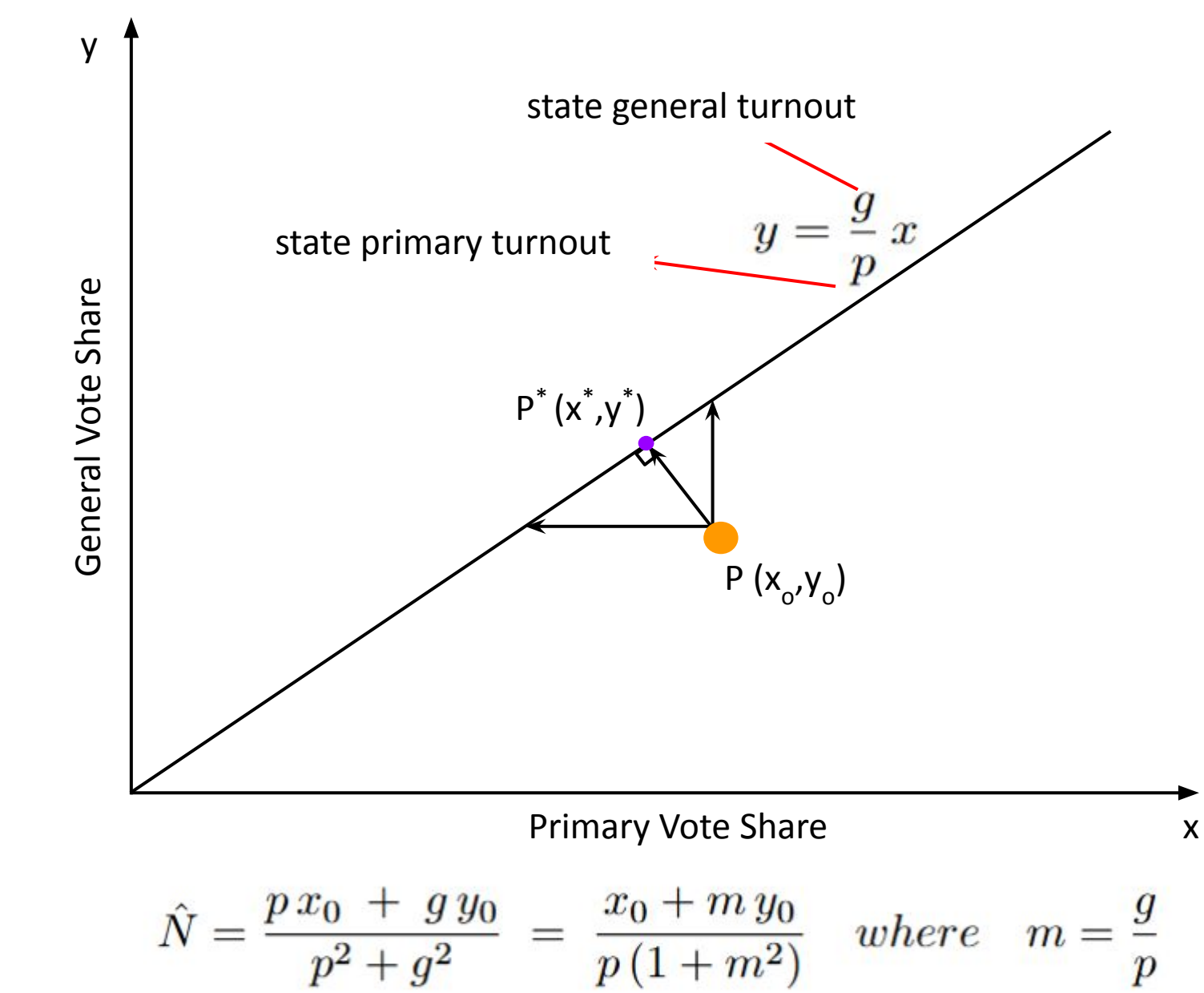
Fig 4b: Probability of voting for registered voters where winning party has lower chance of voting by the result at the state level.

To explain the patterns observed in **Fig. 2** and **Fig. 3**, we propose a **complacency hypothesis**: as the margin in the primary increases, the winning party's voters have a lower probability of turning out in the general election.

$$V = \begin{cases} \frac{p_{\text{winner}} \hat{N}_{\text{winner}}}{p_{\text{winner}} \hat{N}_{\text{winner}} + p_{\text{loser}} (\hat{N} - \hat{N}_{\text{winner}})}, & \text{if } v_{\text{primary}} \geq 0.5, \\ \frac{p_{\text{loser}} \hat{N}_{\text{loser}}}{p_{\text{loser}} \hat{N}_{\text{loser}} + p_{\text{winner}} (\hat{N} - \hat{N}_{\text{loser}})}, & \text{if } v_{\text{primary}} < 0.5. \end{cases}$$

registered voters for winning party $\hat{N}_{\text{winner}} = v_{\text{winner}} \hat{N}$
registered voters for losing party $\hat{N}_{\text{loser}} = v_{\text{loser}} \hat{N}$
 $v_{\text{primary}} = \frac{\text{Voters for party in primary}}{\text{Total primary votes}}$

Estimating Registered Voters



Since the number of registered voters is not directly available at the precinct level, we use state-level turnout probability to estimate number of registered voters by projecting each precinct's data with the state-level turnout probability.

Election modelling result

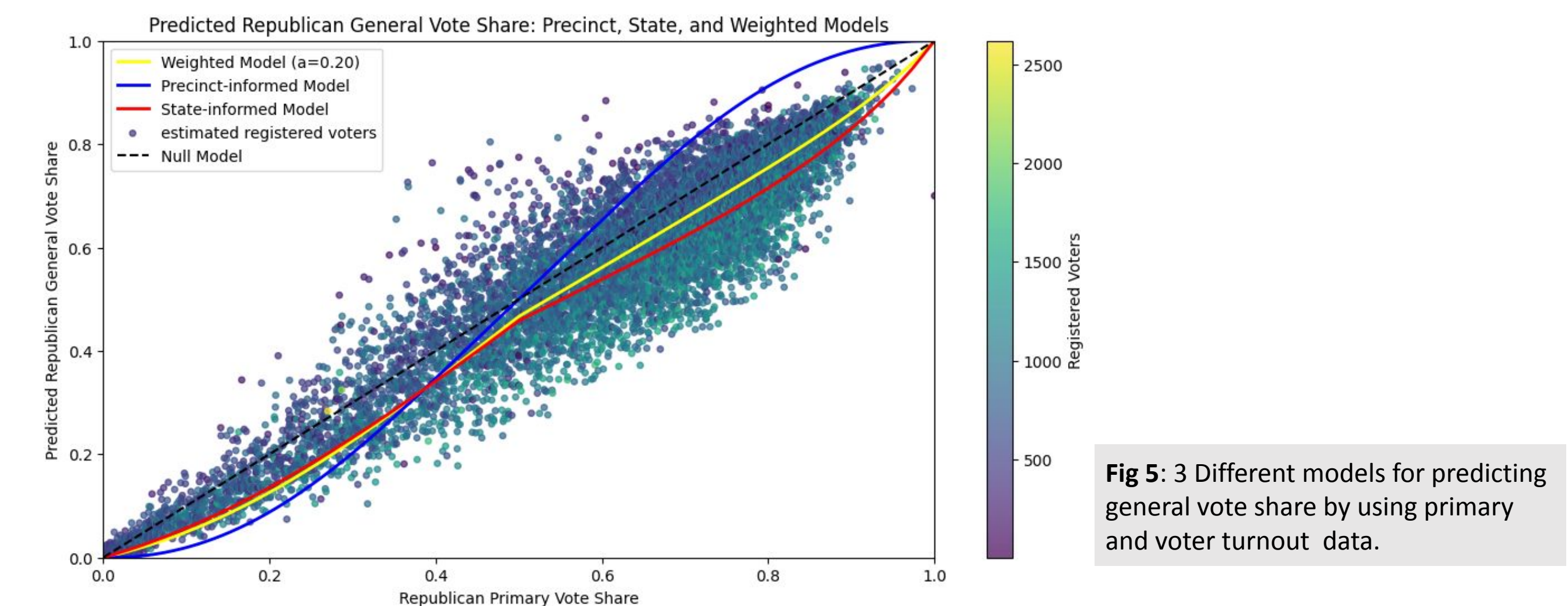


Fig 5: 3 Different models for predicting general vote share by using primary and voter turnout data.

The precinct-informed and state-informed models capture different turnout dynamics, while the weighted model combines both approaches to better match the observed relationship between primary and general vote shares.

Future Steps

Further work is needed to test whether the observed patterns persist across more states and multiple presidential cycles. Because precinct results are spatially correlated, we will develop a **network model** that accounts for geographic proximity between precincts. We also plan to **merge party entropy and candidate ideology** into a single measure of party consensus to better capture its effect on voter turnout and election results.

Acknowledgements

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